

Curricular Vitae

ANWESHA CHAKRABORTY

OrcidID: <https://orcid.org/0000-0002-6110-3770>

Email ID: 24d1261@iitb.ac.in

Education

Sr. No.	Class/ Degree	College/ University	Year of Passing	Marks Obtained	Class/ Division
1.	Class X	Shri Shikshayatan School, West Bengal	2014	610/ 700 (87.14%)	1 st Class
2.	Class XII	St. John's Diocesan Girls' Higher Secondary School, West Bengal	2016	389/ 500 (77.80%)	1 st Class
3.	B.Sc.	Shri Shikshayatan College, West Bengal (University of Calcutta); Botany Honours	2019	529/800 (66.125%)	1 st Class
4.	M.Sc.	Presidency University, West Bengal	2021	CGPA: 8.9/ 10 (83.4%)	1 st Class
5.	PhD	Indian Institute of Technology Bombay	2024- ongoing	GPI: 8.83	na

National Level Examinations Qualified

1. GATE-XL 2024, score- 636/1000, rank- 410
2. 25th State Eligibility Test by West Bengal College Service Commission
3. GATE-XL 2021, score- 394/1000, rank- 2783
4. CSIR-NTA-NET Qualified for Lectureship/ Assistant Professorship, rank- 60 (June-2020)

Publications

1. Chaudhury, R., **Chakraborty, A.**, Rahaman, F., Sarkar, T., Dey, S., & Das, M., **2024**. Mycorrhization in trees: ecology, physiology, emerging technologies and beyond. *Plant Biology*, 26(2), 145-156.
2. Patra, S., **Chakraborty, A.**, & Saikia, P., **2024**. Role of omics tools in understanding the stress tolerance in legumes. In *Current Omics Advancement in Plant Abiotic Stress Biology* (pp. 215-226). **Academic Press**.

3. Basak, P., Dutta, W., Basu, J., Ghosh, M., **Chakraborty, A.**, & Ray, P., **2024**. Phytopathogenic Fungi and Their Active Metabolites with Bioherbicidal Potential Against the Invasive Alligator Weed, *Alternanthera philoxeroides*. **Journal of Phytopathology**, 172(5), e13406.
4. **Chakraborty, A.**, Hussain, A. and Sabnam, N., **2023**. Uncovering the structural stability of *Magnaporthe oryzae* effectors: a secretome-wide in silico analysis. **Journal of Biomolecular Structure and Dynamics**, pp.1-22.
5. **Chakraborty, A.** and Saikia, P., **2023**. Climate Change Impacts on the Higher Altitude Forests of Indian Himalayan Regions: Nature-Based Solutions for Climate Resilience and Disaster Risk Reduction. In The Palgrave Handbook of Socio-ecological Resilience in the Face of Climate Change: Contexts from a Developing Country (pp. 313-328). Singapore: **Springer Nature Singapore**.
6. Tyagi, S., Yadav, P., **Chakraborty, A.**, Majumdar, A., Moulick, D., Santra, S.C., Upadhyay, M.K., Sahoo, U., Maitra, S. and Hossain, A., **2023**. Myconanotechnologies: an approach towards sustainable agriculture. In Myconanotechnology and Application of Nanoparticles in Biology (pp. 139-157). **Academic Press**.
7. **Chakraborty, A.**, Chaudhury, R., Dutta, S., Basak, M., Dey, S., Schäffner, A.R. and Das, M., **2022**. Role of metabolites in flower development and discovery of compounds controlling flowering time. **Plant Physiology and Biochemistry**.
8. Yadav, P., **Chakraborty, A.**, Srivastava, S., Sahani, S. and Singh, P., **2022**. Phytoremediation: A sustainable technology for pollution control and environmental clean-up. In Innovative Bio-Based Technologies for Environmental Remediation (pp. 237-249). **CRC Press**. eBook ISBN: 9781003004684. Year: 2022
9. **Chakraborty, A.** and Ray, P., **2021**. Mycoherbicides for the Noxious Meddlesome: Can *Colletotrichum* be a Budding Candidate? **Frontiers in Microbiology**, 12, p.754048.

PhD work and thesis statement

Research Aims: The primary aim is material synthesis (biochar) via thermal degradation/ slow pyrolysis using aquatic weeds or other agricultural-wastes and apply them for as soil amendment to study its impact on crop growth. The same phenotypic effect will be further traced down to the genotypic (developmental gene) level to understand how growth promotion by biochar application is affecting at the gene level. Following this the native soil microbial system under impact of biochar will also be studied. It is hypothesised that a microbial community shift may be observed under prolonged exposure to biochar in the soil system. This interaction shall be studied at two levels: (i) community shift at genus and species level and (ii) quantum level change involving electronic conversation (Extracellular Electron Transfer) between biochar and the inhabitant microbes.

Current work: My latest work has shown biochar from aquatic weed shows positive impact on mustard plant growth. This work is currently being prepared for a publication. Currently I have started utilizing DFT-based studies to simulate the electronic communication signatures that facilitate interactions at the carbon (on biochar) and microbe (cell wall/ protein structures) interface.

Job/ Experience

Project Title: 'Functional characterization of novel secreted effectors in *Magnaporthe oryzae*'.

- **Details:** Funded by SERB-DST, Government of India, India. Project supervisor: Dr. Nazmiara Sabnam [Assistant Professor, Presidency University, Kolkata]. Internship project, with fellowship.
- **Job description:** I undertook following studies under the above-mentioned topic.

a) Understanding the extent of intrinsic disorder in pathogenic effector proteins.

In this study around 365 small secretory effector molecules were bioinformatically analysed from the entire secretome of rice-blast pathogen *Magnaporthe oryzae*, essential characteristics like presence of a signal peptide, absence of any transmembrane domain, signal peptide cleavage site, etc were also bioinformatically analysed for all the identified effectors. An intricate public bioinformatic tool 'FoldIndex' was used to know some aspects like 'extent of intrinsic disorder', 'extent of infoldability', 'charge', 'phobicity' of all the effectors. Further, obtained results were analysed and the effector proteins were classified as 'ordered', 'mostly ordered', 'moderately disordered', 'mostly disordered', 'highly disordered' and 'extremely disordered'.

b) Understanding some crucial aspects of pathogenic effector molecules at molecular level.

This study was part of a bigger project. Some crucial aspects like natural intrinsically disordered regions, percentage of protein disorder, sites of phosphorylation in the amino acid sequences, possible domains and their functions, and post translational modification sites were analyzed in three hypothetical effector genes in *Magnaporthe oryzae*.

Projects (experiences)

Promoter analysis of osmotic stress responsive genes in representative species from Brassicaceae.

- **Details:** MS dissertation project. Associated with Presidency University, Kolkata, laboratory of plant molecular biology, supervisor: Dr. Malay Das Sir [Assistant Professor, Presidency University, Kolkata].
- **Work Description:** An intricate bioinformatic analysis was performed to study four gene families (*DREB1*, *SWEET*, *LIM*, *ANNEXIN*) known to have stress responsive characters in Brassicaceae. Model study plants were *Thellungiella salsuginea* (highly stress tolerant species) and *Arabidopsis thaliana* (stress sensitive species).

Certificates obtained (selected ones)

1. **Best Oral Presentation Award** at **International Conference on Energy and Resource Recovery for a Sustainable Circular Economy, 2026** held at **IIT Madras**.

2. **Certificate of Merit for Securing First Class in Botany Honours** from Shri Shikshayatan College, 29.09.2019
3. **Certificate of Excellence in Academic Pursuits** from Shri Shikshayatan College, 06.07.2019.
4. **Best Poster (2nd Position) Certificate Cum Award in State Level Seminar** on Topic Conservation of Biodiversity: Impact, Challenges and Management Organised by Shri Shikshayatan College, 06.04.2019.
5. **Certificate of Oral Presentation at State Level Workshop**, Topic Role of Youth in Solid Waste Management, Organised by Shri Shikshayatan College in Collaboration with **West Bengal Pollution Control Board**, 07.11.2017.
6. Other Certificates of Appreciation in the field of Art, Dance and Extra-Curricular Activities.

Awards/ Medals

1. Medal for obtaining **First Class** in B.Sc. Botany Honours.

Extra-Curricular Courses/ Activities

1. Cultural Secretary of Hostel 10, Indian Institute of Technology Bombay (2025- 2026).
2. Design Team Head of Bengali Cultural Association of Indian Institute of Technology Bombay (2025-26).
3. Core Management Team Member of W-CARESOL workshop at Indian Institute of Technology Bombay.
4. British Council Course- Cambridge Young Learners English Tests Movers and Flyers Qualified.
5. Yoga. Painting. Classical Dance. Swimming. Karate. Camping/ Trekking.

Membership

1. Associate member of Young Academy of India (Active member, since 2020)

Self-declaration: I acknowledge that the above information is true to my knowledge.

Anwesha Chakraborty